

Ramcharan Rajpurohit

+91-9680103852 | ramcharanrajpurohit@gmail.com | b23ci1032@iitj.ac.in

GitHub | LinkedIn | Website | Codolio

EDUCATION

IIT Jodhpur

Bachelor of Technology

Sep 2023 – May 2027

Jodhpur, Rajasthan

WORK EXPERIENCE

- **AI Engineering Intern at PowersMy.biz** [\[Live Platform\]](#) | Nov 2025 – Jul 2026
 - Built a production AI assessment platform for JEE and NEET coaching institutes serving **1,000+ concurrent exam-takers** (FastAPI, LangGraph, React, Flutter, MongoDB, Redis, GCP).
 - **Agentic AI (Amigo)**: architected a student-facing AI companion on LangGraph, where a supervisor graph routes to **5 specialist agents** over **18 scoped tools** and serves text chat and voice from one shared graph.
 - **Realtime Voice**: implemented Amigo's LiveKit voice pipeline with code-switching Hindi and English speech recognition, streaming synthesis, and barge-in handling, speaking each token as the model produces it.
 - **AI Safety & Access Control**: gated every agent data access behind a **scope-checked broker** with an append-only audit ledger, screening each turn with PII tokenisation and a fail-closed risk classifier.
 - **Multimodal AI**: engineered an evaluation pipeline using vision models to grade handwritten subjective answers across 8 question formats and identify learning gaps.
 - **Backend Performance**: optimized high-concurrency exam APIs with Redis caching, atomic MongoDB operations, and off-thread bcrypt processing to sustain peak load.
 - **Distributed Systems**: designed an asynchronous grading queue across 2 services returning objective scores instantly while written submissions evaluate in the background.
 - **Cloud & DevOps**: led GCP deployment of the FastAPI backend (Cloud Run, VPC connector, auto-scaling, concurrency tuning) to handle exam-day traffic spikes.
 - **Document AI**: shipped a pipeline turning exam PDFs into a structured question bank, running **Mistral OCR** and a **DocLayout-YOLO** figure detector in parallel with per-subject Gemini extraction.

PROJECTS

- **DSA Visualizer** – Interactive Algorithm Learning Platform [\[Live Site\]](#) | [\[GitHub\]](#)
 - Developed a framework-free algorithm visualizer in raw ES6 and HTML5 Canvas, animating 15+ sorting, graph, and tree algorithms step by step.
 - Architected a class-based OOP rendering engine where each algorithm is a self-contained plug-in subclass, with frame-by-frame controls for recursion tracing and backward stepping.
- **DevPlaza** – Competitive Programming Profile Aggregator [\[Live Site\]](#) | [\[GitHub\]](#)
 - Consolidated stats from 5 coding platforms (Codeforces, LeetCode, GeeksforGeeks, CodeChef, AtCoder) into one MERN dashboard, pulling undocumented data from private GraphQL and REST endpoints using rotating user agents and custom headers.
 - Hardened every feed with retry logic and fallback parsers for upstream outages, and secured the app with 3 sign-in methods (JWT, Google OAuth, email OTP).

TECHNICAL SKILLS

- **AI/ML & GenAI**: LangGraph, LangChain, Multi-Agent Systems, Gemini API, OpenAI API, Vision Models, Realtime Voice Agents (LiveKit, STT, TTS), Prompt Engineering
- **Programming Languages**: C++, C, Python, JavaScript, TypeScript
- **Frontend Development**: React.js, HTML5, CSS3, TailwindCSS, Redux, Flutter
- **Backend Development**: Node.js, Express.js, FastAPI, REST APIs, GraphQL, SSE Streaming, WebRTC, JWT, OAuth 2.0, MongoDB, Redis
- **Cloud & Tools**: GCP (Cloud Run, VPC), Docker, GitHub Actions, Git, Postman, Linux, Vercel, CI/CD

ACHIEVEMENTS

- Authored a **Linux kernel** patch merged into mainline, fixing a broken mic-mute LED on a personal HP laptop after root-causing the missing audio driver quirk. [\[Commit\]](#)
- Selected for **Amazon Summer School 2025**.
- Achieved **2nd District Rank** in Class 12 RBSE Board Exams with a 98.20% aggregate.

LEADERSHIP

- Core Member, Dramatics Society (**Dramebazz**), IIT Jodhpur; coordinated events and creative direction, and directed stage productions at **Inter IIT Cultural Meet 7.0**.